

NANTUCKET BUOY 44008, USA

WMO: 992790

Lat: 40.504N

Lon: 69.248W

Elev: 0

StdP: 14.70

Time Zone: -5.00 (W05)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	21.8	25.2	12.0	10.2	25.3	15.1	11.8	27.0	40.4	34.3	38.8	34.8	23.0	320	0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.6	75.3	73.4	74.2	72.3	73.0	71.2	74.6	75.1	73.4	74.0	72.2	73.1	12.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.4	128.6	74.9	73.2	123.2	73.9	71.9	117.7	72.8	38.1	75.5	37.0	74.2	35.9	73.1	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.5	33.4	30.5	DB	19.6	77.1	4.0	1.6	16.7	78.2	14.3	79.1	12.0	80.0	9.1	81.2	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.3	39.3	38.7	40.4	45.2	51.0	57.8	65.7	68.1	65.3	58.0	51.1	45.5
	DBStd	11.37	7.49	6.17	4.72	3.37	3.59	4.20	3.73	3.61	4.05	4.72	5.81	6.34
	HDD50	1354	334	318	299	147	30	0	0	0	0	3	55	168
	HDD65	4877	796	736	763	593	434	218	37	14	44	221	416	605
	CDD50	2180	3	2	0	3	61	235	486	562	460	251	89	28
	CDD65	229	0	0	0	0	0	2	58	111	54	4	0	0
	CDH74	104	0	0	0	0	0	0	16	59	29	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	14.5	18.9	18.8	17.0	14.8	11.9	10.3	9.3	9.7	12.0	15.7	17.4	18.5
Precipitation	PrecAvg	59.2	5.6	5.3	5.2	4.6	4.1	3.7	3.9	4.8	4.9	5.0	5.7	6.6
	PrecMax	85.3	8.6	8.2	9.1	7.8	8.9	8.1	7.0	18.1	9.9	10.4	10.4	9.1
	PrecMin	49.5	3.2	2.4	2.4	1.6	1.3	0.6	1.7	1.0	1.3	1.7	1.9	3.5
	PrecStd	7.7	1.4	1.5	1.5	1.4	1.5	1.9	1.5	3.3	2.4	2.1	1.7	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.1	56.8	52.1	55.0	62.9	69.3	75.9	77.0	76.4	71.3	65.7	61.0
		MCWB	55.3	55.2	51.2	53.5	62.3	66.4	74.4	74.6	74.6	70.0	63.0	59.9
	2%	DB	53.5	51.6	50.4	53.0	60.0	67.4	74.1	75.6	74.9	68.6	62.5	58.5
		MCWB	52.1	50.5	49.5	51.8	59.4	66.1	72.4	73.4	73.2	67.0	60.4	56.9
	5%	DB	51.7	49.7	49.1	51.7	57.9	65.8	72.7	74.6	73.3	66.3	60.6	56.8
		MCWB	50.4	48.4	48.0	50.7	57.1	64.7	70.9	72.6	71.8	64.2	58.6	55.1
	10%	DB	49.7	47.3	47.3	50.4	56.2	64.3	71.3	73.4	71.8	64.4	59.0	55.0
		MCWB	48.2	45.8	45.8	49.1	55.1	63.3	69.8	71.6	69.6	62.3	57.2	53.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.2	52.4	51.9	54.2	62.9	67.5	75.2	75.8	76.2	69.5	64.3	61.0
		MCDB	56.5	52.8	52.2	54.6	63.3	67.8	75.6	76.2	76.2	69.5	65.0	61.2
	2%	WB	53.8	49.9	50.3	52.4	60.0	66.0	73.5	74.8	74.2	67.0	62.1	57.9
		MCDB	54.3	50.5	50.6	52.8	60.3	66.6	74.1	75.3	74.6	67.4	63.4	58.5
	5%	WB	51.5	47.6	48.8	51.2	57.9	64.7	72.0	73.8	72.4	65.2	59.8	56.1
		MCDB	52.4	48.6	49.3	51.7	58.1	65.4	72.6	74.4	73.4	65.9	60.7	57.1
	10%	WB	49.7	45.4	46.7	49.8	55.9	63.3	70.6	72.8	70.7	63.3	58.2	54.4
		MCDB	50.7	46.6	47.7	50.4	56.3	64.0	71.4	73.6	72.0	64.5	59.3	55.7
Mean Daily Temperature Range	5% DB	MDBR	8.5	8.2	6.5	4.9	4.8	5.1	5.8	5.6	4.9	5.2	6.0	8.0
		MCDBR	10.9	11.0	8.8	7.1	6.4	6.0	6.5	6.1	6.6	7.1	7.9	10.1
		MCWBR	12.0	11.8	9.5	7.2	5.9	5.8	5.9	5.6	6.7	8.2	9.0	11.8
	5% WB	MCDBR	10.3	10.6	8.4	7.0	6.0	6.1	6.1	5.6	6.6	6.4	7.4	10.3
		MCWBR	11.9	11.7	9.8	7.4	5.9	5.8	5.9	5.4	6.9	8.0	8.4	11.8
Clear-Sky Solar Irradiance	taub	0.340	0.347	0.359	0.389	0.424	0.455	0.480	0.443	0.383	0.372	0.342	0.332	
	taud	2.433	2.419	2.426	2.358	2.297	2.248	2.196	2.303	2.433	2.462	2.511	2.487	
	Ebn at Noon	258	274	283	280	272	263	254	261	272	262	257	252	
	Edh at Noon	26	30	33	38	41	43	45	40	33	29	24	23	
All-Sky Solar Radiation	RadAvg	501	797	1197	1569	1784	1951	1951	1792	1472	1009	621	441	
	RadStd	45	77	105	156	182	155	112	122	122	75	69	39	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+155	N/A

Nomenclature: See separate page